4.1.1 Cellular respiration

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	an outline of glycolysis as an enzyme controlled metabolic pathway	To include the location of the enzymes (only dehydrogenase to be named), the conversion of glucose to triose phosphate, the production of pyruvate, reduced NAD and the production of ATP by substrate level phosphorylation. HSW8
(b)	an outline of the link reaction	To include the location of the enzymes, decarboxylation of pyruvate (3C) to acetyl (2C) coenzyme A and the reduction of NAD. HSW8
(c)	an outline of the Krebs cycle	To include the location of the enzymes, the formation of citrate from acetate and oxaloacetate, the conversion of citrate to oxaloacetate using decarboxylation and dehydrogenation reactions and the production of reduced NAD and reduced FAD and also ATP by substrate level phosphorylation (names of intermediate compounds are not required). HSW8

(d)	an outline of the process of oxidative phosphorylation	To include the location of the process and the roles of electron carriers, oxygen and the mitochondrial cristae (inner membrane) **AND** to include the electron transport chain, proton gradients and ATP synthase. HSW8
(e)	an outline of the process of anaerobic respiration in muscle cells and in yeast	To include an appreciation that anaerobic respiration produces a much lower yield of ATP than aerobic respiration.
(f)	the relative energy values of different respiratory substrates	To include the calculation and interpretation of respiratory quotients for carbohydrates, proteins and lipids. *M0.3, M2.3, M2.4* HSW8, HSW12
(g)	**(i)** the use of respirometers and other methods to investigate the rate of respiration **(ii)** practical investigations into the effect of temperature, substrate concentration, anaerobic conditions and different respiratory substrates on the rate of respiration.	*M0.1, M0.2, M0.3, M1.1, M1.2, M1.3, M1.6, M1.9, M3.2, M3.3, M3.5, M3.6* **PAG4** HSW2, HSW3, HSW4, HSW5, HSW6